The Silicon Chokepoint

Taiwan, TSMC, and the Geopolitics of Advanced Semiconductor Manufacturing

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The global production of advanced semiconductor chips is more geographically concentrated than almost any other critical industrial input. Petroleum reserves are distributed across dozens of countries on six continents. Rare earth elements, despite the name, are mined on four continents. Advanced logic chips – the class of semiconductor that powers AI accelerators, cloud computing infrastructure, and modern weapons guidance systems – are produced at leading-edge process nodes in meaningful volume by exactly one company operating on one island 110 miles from the Chinese mainland.

Taiwan Semiconductor Manufacturing Company accounts for approximately 90 percent of global production of chips manufactured at process nodes below 10 nanometers.[1] This concentration is not incidental. It reflects decades of accumulated process engineering expertise, a dense cluster of specialized suppliers and equipment vendors, a deep pool of trained engineering labor, and returns to scale that make geographic replication difficult under any conditions. It also means that any significant disruption to TSMC's operations – whether from natural disaster, political crisis, or military conflict – would remove from the global economy a productive capacity that cannot be replaced in the short or medium term.

This paper examines the foreign policy implications of that concentration: what it means for the US-China rivalry over technology dominance, what the economic literature says about the risks of strategic dependence on single-source critical inputs, and how the United States and its partners have begun to respond.

The Structure of the Dependence

TSMC's dominance at advanced process nodes is the product of a specific historical trajectory. The company's founder, Morris Chang, established TSMC in 1987 on the premise of a pure-play foundry model: TSMC would manufacture chips designed by others, investing in fabrication capability without competing with its customers on chip design. This model attracted the global fabless semiconductor industry – Nvidia, Qualcomm, Apple, AMD, Broadcom – which could specialize in chip architecture without building their own fabs. The resulting volume of diverse advanced orders gave TSMC the capital and the learning curve advantages to sustain continuous process node advancement, which in turn attracted more customers, which generated more revenue and learning.[2]

By 2022, TSMC's revenue from customers in North America alone exceeded $35 billion annually. Its advanced node capacity was the linchpin of the global technology supply chain in a way that became visible to policymakers only when the COVID-19 pandemic produced chip shortages that idled automobile assembly lines, delayed consumer electronics launches, and prompted a White House supply chain review.[3]

The review's findings, published in June 2021 across four sector-specific reports, documented not only the concentration of advanced semiconductor manufacturing in Taiwan but the extent to which U.S. firms had offshored fabrication over the preceding three decades. In 1990, the United States had produced approximately 37 percent of global semiconductor output by value. By 2020, that share had fallen to 12 percent, and essentially none of the leading-edge production.[4] The reports framed this as both an economic vulnerability and a national security risk, noting that U.S. military systems depended on chips fabricated abroad and that a sustained disruption to Taiwan's production would impose costs that no domestic alternative could quickly absorb.

The Taiwan Contingency and Its Economic Weight

The economic cost of a significant disruption to Taiwanese semiconductor production is among the most studied supply chain risks in current economic analysis. A 2021 study by the Boston Consulting Group and the Semiconductor Industry Association estimated that a one-year disruption to Taiwanese chip supply would produce a direct revenue loss to downstream electronics industries of approximately $490 billion annually, with cascading effects on sectors from automotive to defense that are harder to quantify but likely comparable in magnitude.[5]

These figures reflect the absence of short-run substitutes. Semiconductor fabrication at leading-edge nodes requires highly specialized equipment, ultra-pure materials, and process engineering knowledge that accumulates over years of production experience. A new fab building from greenfield to volume production at an advanced node requires three to five years and capital expenditures in the range of $15 to $20 billion.[6] Even with unlimited capital, the process engineering knowledge needed to achieve competitive yield at a new node is not transferable by blueprint; it must be developed through iterative production experience. A disruption to TSMC's operations would therefore produce an output gap that no actor could close within a timeframe relevant to acute geopolitical crises.

The Taiwan contingency has particular weight in the context of US-China strategic competition because it creates a structural asymmetry: the United States and its allies depend on Taiwanese production for the chips that underpin their AI development programs, their defense systems, and their commercial technology industries, while China's stated foreign policy objective includes eventual political unification with Taiwan under terms that Chinese leaders have declined to commit to being peaceful. The supply chain vulnerability and the political risk are not independent; they are coupled in a way that makes the geographic concentration of chip production a foreign policy problem rather than merely a commercial one.

The Silicon Shield Hypothesis and Its Limits

One argument that circulates in Taiwan policy discussions holds that TSMC's economic importance itself provides a deterrent against Chinese military action – that the world's dependence on Taiwanese chips creates a coalition of interested parties whose economic stakes in Taiwan's continued functioning exceed their willingness to absorb the cost of disruption. This argument, sometimes called the "silicon shield" hypothesis, has intuitive appeal but several structural weaknesses.

The deterrent effect requires that potential aggressors value the economic consequences of disruption enough to be deterred by them. A government willing to accept significant economic cost in pursuit of a territorial objective is not deterred by the prospect of that cost. The silicon shield hypothesis also assumes that Taiwan's chip production would be preserved through a conflict, which depends on the character of the conflict and on TSMC's response to it. TSMC's management has publicly stated that the company's fabs would be unworkable without continuous global supply of equipment, chemicals, and software – implying that an effective blockade, rather than direct destruction, could degrade TSMC's output without physically damaging the facilities.[7]

The limits of the silicon shield argument do not mean that Taiwan's economic importance provides no deterrent effect; they mean that it is not a sufficient deterrent by itself, and that US foreign policy cannot treat Taiwan's semiconductor concentration as a substitute for conventional deterrence.

The US Policy Response

The United States has pursued two parallel responses to the strategic dependence on Taiwanese semiconductor production, with different timeframes and different mechanisms.

The first is short-run: export controls designed to prevent China from developing a domestic advanced semiconductor capability that would reduce its dependence on the global supply chain while enabling military applications. The October 2022 Bureau of Industry and Security rules, and their October 2023 successor, represent the most aggressive use of technology export controls since the Cold War. Their effectiveness depends on allied coordination, particularly with the Netherlands (home to ASML, the sole supplier of EUV lithography) and Japan (home to several critical equipment and materials suppliers), both of which implemented coordinated restrictions.[8]

The second is long-run: domestic capacity investment. The CHIPS and Science Act, signed in August 2022, appropriated $52.7 billion for domestic semiconductor research, development, and manufacturing, with $39 billion specifically allocated for manufacturing incentives.[9] The legislation was the direct legislative response to the supply chain vulnerability documented in the 2021 White House review. It attracted TSMC's commitment to build two advanced fabrication facilities in Arizona – a 4nm facility targeting production in 2024 and a 2nm facility announced in 2023 – as well as investments by Intel, Samsung, and Micron at domestic sites.

Japan's parallel response, a government-backed initiative called Rapidus targeting domestic 2nm chip production by 2027 in partnership with IBM, and TSMC's construction of a fab in Kumamoto with Japanese government support, represent allied efforts toward the same goal of reducing single-point supply chain vulnerability.[10]

What Geographic Diversification Can and Cannot Accomplish

The policy responses above are meaningful, but the economics of semiconductor manufacturing impose limits on what geographic diversification can achieve within any medium-term timeframe.

TSMC's competitive advantage over potential rivals rests not only on its equipment and facilities but on the density of its supplier ecosystem in Taiwan, the size and depth of its engineering workforce, and the accumulated process engineering knowledge embedded in its production systems. An Arizona fab staffed partly by engineers trained in Taiwan and supplied by a global equipment chain is less vulnerable to Taiwan-specific disruption than production concentrated entirely on the island, but it does not replicate the full economic environment that makes TSMC productive. TSMC's management acknowledged this explicitly: the company's Arizona fabs are estimated to produce chips at a cost premium of 50 percent or more compared to equivalent production in Taiwan, reflecting higher labor costs, less mature local supplier ecosystems, and the absence of the cluster efficiencies that characterize the Hsinchu and Tainan science parks.[11]

The policy implication is that domestic capacity investment reduces but does not eliminate strategic vulnerability to Taiwan disruption. It also creates a new dynamic in the US-China competition: to the extent that U.S. policy succeeds in widening China's chip capability gap while diversifying Western production away from Taiwan, it simultaneously reduces China's ability to leverage the supply chain dependence as a coercive instrument.

References

[1] Boston Consulting Group and Semiconductor Industry Association, "Strengthening the Global Semiconductor Supply Chain in an Uncertain Era" (April 2021), p. 4. The sub-10nm figure reflects TSMC's dominant share of advanced foundry production; Samsung accounts for most of the remainder at advanced nodes.

[2] Chris Miller, *Chip War: The Fight for the World's Most Critical Technology* (New York: Scribner, 2022), chapters 14-16. Miller's account of TSMC's foundry model and its consequences for global industry structure is the most comprehensive available treatment.

[3] The White House, *Building Resilient Supply Chains, Revitalizing American Manufacturing, and Fostering Broad-Based Growth: 100-Day Reviews under Executive Order 14017* (June 2021).

[4] Semiconductor Industry Association, "State of the U.S. Semiconductor Industry 2021" (2021), p. 6. The 37 percent to 12 percent decline is a frequently cited figure in CHIPS Act legislative history and is drawn from SIA data.

[5] Boston Consulting Group and Semiconductor Industry Association (April 2021), p. 18. The $490 billion figure represents estimated annual revenue losses to downstream electronics industries; the report notes that indirect effects would be substantially larger.

[6] U.S. Department of Commerce, Bureau of Industry and Security, "Assessment of the Critical Supply Chains Supporting the U.S. Information and Communications Technology Industry" (February 2022), pp. 14-16.

[7] TSMC CEO C.C. Wei, comments at TSMC's 2022 Annual General Meeting, as reported by Reuters (June 2022). Wei stated that TSMC's operations require continuous global supply of several thousand specialized inputs and that the fabs would not be operable if that supply were severed.

[8] Dutch Ministry of Foreign Affairs export license policy update for ASML (January

2023); Japan Ministry of Economy, Trade and Industry semiconductor equipment export control amendments (March 2023).

[9] CHIPS and Science Act of 2022, P.L. 117-167 (August 9, 2022). The manufacturing incentive allocation and the research and workforce development allocations are detailed in Division A (CHIPS for America) of the Act.

[10] Rapidus Corporation, established October 2022 with backing from Toyota, Sony, NTT, and the Japanese government; TSMC Kumamoto Fab 1 groundbreaking, November 2022. See Ministry of Economy, Trade and Industry of Japan press releases, November 2022.

[11] TSMC management commentary at Q4 2022 earnings call (January 2023), in which the company discussed cost differentials between Arizona and Taiwan production and attributed them to labor market, supplier ecosystem, and infrastructure differences. The 50 percent premium estimate reflects subsequent analyst assessments and is directionally consistent with TSMC's own public characterizations.